

-- Exercise 1: Creating a Non-Clustered Index

-- Goal: Create a non-clustered index on the ProductName column in the Products table and compare query execution time before and after index creation.

-- Step 1: Query to fetch product details before index creation

```
SELECT * FROM Products WHERE ProductName = 'Laptop';
```

-- Step 2: Create a non-clustered index on ProductName

```
CREATE NONCLUSTERED INDEX IX_Products_ProductName  
ON Products(ProductName);
```

-- Step 3: Query to fetch product details after index creation

```
SELECT * FROM Products WHERE ProductName = 'Laptop';
```

-- Exercise 2: Creating a Clustered Index

-- Goal: Create a clustered index on the OrderDate column in the Orders table and compare query execution time before and after index creation.

-- Step 1: Query to fetch orders before index creation

```
SELECT * FROM Orders WHERE OrderDate = '2023-01-15';
```

-- Step 2: Create a clustered index on OrderDate

```
CREATE CLUSTERED INDEX IX_Orders_OrderDate  
ON Orders(OrderDate);
```

-- Step 3: Query to fetch orders after index creation

```
SELECT * FROM Orders WHERE OrderDate = '2023-01-15';
```

-- Exercise 3: Creating a Composite Index

-- Goal: Create a composite index on the CustomerID and OrderDate columns in the Orders table and compare query execution time before and after index creation.

-- Step 1: Query to fetch orders before index creation

```
SELECT * FROM Orders WHERE CustomerID = 1 AND OrderDate = '2023-01-15';
```

-- Step 2: Create a composite index on CustomerID and OrderDate

```
CREATE INDEX IX_Orders_CustomerID_OrderDate  
ON Orders(CustomerID, OrderDate);
```

-- Step 3: Query to fetch orders after index creation

```
SELECT * FROM Orders WHERE CustomerID = 1 AND OrderDate = '2023-01-15';
```